Superstore Sales — Business Report

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Period covered: January 2014 – December 2017 (9,994 order lines) **Prepared from:** `Superstore sales dataset.csv` **Supporting detail:** see `outputs/RESULTS.md` for every chart and the numbers behind this report

Executive summary

Superstore grew sales every year over the four-year period, with a strong and consistent year-end peak. But revenue growth is masking a profitability problem concentrated in a few, fixable places: one region, one product category, and a discounting practice that turns roughly one in five order lines into a loss.

- **Discounting is the single biggest driver of unprofitability.** Order lines that lose money carry an average discount of 48%, versus 8% on profitable lines. Profit margin and discount rate correlate at -0.86 — among the strongest relationships in the data.
- **Furniture is structurally unprofitable at scale.** The category converts only 2.5% of sales to profit, against 17% for Office Supplies and Technology. Tables alone gave back $17,725 in the period — the single largest profit drag in the business — despite being a top-6 product by sales volume.
- **Central region underperforms on margin, not volume.** It converts under 8% of sales to profit versus 15% in the West, despite not being the smallest region by revenue.
- **A 12-month forecast projects $778,333 in sales for 2018**, 6.2% above the trailing 12 months, consistent with the historical growth trend and the recurring Q4 peak.

Key findings

1. Profitability is uneven across region and category

Cut	Best	Worst	Gap
Region (margin)	West, 14.9%	Central, 7.9%	~2x
Category (margin)	Technology, 17.4%	Furniture, 2.5%	~7x
Segment (margin)	Home Office, 14.0%	Consumer, 11.5%	—

Central and Furniture are not small — they are meaningful revenue lines that happen to convert poorly to profit. That makes them a margin-improvement target, not a "shrink the business" problem.

2. Discounting is eroding — and in many cases erasing — margin

- Profit margin falls almost linearly as discount increases, and turns negative around a 20% discount for Furniture specifically; Technology and Office Supplies tolerate deeper discounts before turning negative, but all three categories converge around a **-120% margin** at discounts of 51% or more.
- **1,871 of 9,994 order lines (18.7%) are unprofitable**, giving back $156,131 in total profit. These lines are not random — they carry roughly 6x the average discount rate of profitable lines.
- **Tables, Bookcases, and Supplies** are the only three sub-categories (of 17) that lose money overall.

This is the highest-leverage finding in the dataset: it is a pricing/discount governance issue, not a demand issue. The products and regions involved still sell well.

3. Geography: strong revenue states are not always profitable states

California and New York are both the largest markets by sales and healthy contributors to profit. **Texas is the exception** — the #3 state by sales revenue but the single largest loss-maker overall (-$25,729), followed by Ohio, Pennsylvania, Illinois, and North Carolina, all net-negative on profit despite solid sales volume. This points to a regional discounting or account-level pricing practice worth auditing specifically in these states.

4. Shipping delay is a customer/ship-mode choice, not a fulfillment problem

Average delay ranges from same-day (0 days) to Standard Class (5 days), and that range is almost entirely explained by which ship mode was selected — average delay barely moves across regions (3.9–4.1 days). There is no evidence of a regional logistics bottleneck. Separately, **customers are steadily trading up to faster shipping**: Standard Class's share of orders fell from 62% to 57% between 2014 and 2017 while First Class grew from 15% to 17%, a trend worth watching for its effect on fulfillment cost.

5. Seasonality is strong, consistent, and plannable

September, November, and December are consistently the three strongest months every year; January and February are consistently the weakest. This pattern repeats reliably enough across all four years to drive inventory and staffing plans directly, rather than treating each year-end surge as unexpected.

6. Revenue concentration is real but not extreme

The top 10% of customers (of 793 total) drive 31% of sales; the top 20% drive 48%. This is a healthy, broad-based customer file rather than a business dependent on a handful of accounts — retention efforts should target the top quintile as a cohort rather than a short list of named accounts.

7. Order values are highly right-skewed

Median order value is $152; the mean ($459) is pulled well above it by a small number of large orders (up to $23,661). Any revenue-per-order reporting to leadership should lead with the median, not the mean, to avoid overstating the typical transaction.

Forecast and outlook

Monthly sales were split into a 42-month training window and a 6-month holdout. Two forecasting approaches were evaluated on the holdout:

Model	MAPE	Verdict
Prophet	20.7%	**Selected**
SARIMA (auto-ARIMA)	27.8%	Runner-up

Prophet was refit on the full four-year history and projected forward 12 months. It forecasts **$778,333 in sales for January–December 2018**, a 6.2% increase over the trailing 12 months, with the familiar Q4 peak reappearing on schedule.

How to use this forecast: a ~21% MAPE means the forecast is reliable for directional planning — budgeting, staffing, inventory build-up ahead of Q4 — but individual months carry real uncertainty (see the confidence band in `outputs/charts/16_forecast.png`) and it should not be read as a precise monthly revenue commitment. The model was trained only on historical sales patterns; it does not account for planned promotions, pricing changes, or macroeconomic shifts, so it should be refreshed as new months of actuals land.

Recommendations

1. **Cap or restructure discounting on Furniture**, particularly Tables and Bookcases. Margin turns negative well before the discount levels currently in use; a lower discount ceiling or a pricing floor by sub-category would directly address the largest profit drag in the business.
2. **Audit pricing/discount practice in Texas, Ohio, Pennsylvania, Illinois, and North Carolina.** These states combine strong sales with net losses, suggesting a local or account-level discounting pattern rather than weak demand.
3. **Investigate Central region's margin gap** relative to West/East — same playbook as the Furniture issue, applied geographically.
4. **Plan inventory and staffing around the Sep–Dec peak** and the Jan–Feb trough; the pattern is consistent enough across four years to commit to, not just react to.
5. **Monitor the shift toward First Class/Same Day shipping** as a fulfillment- cost trend, and consider whether shipping is priced (or subsidized) correctly for the mode mix customers are actually choosing.
6. **Use the 2018 forecast for budget planning**, refreshing the model quarterly as actuals arrive, and treat single-month forecasts as a range rather than a point estimate.

Methodology and data notes

- Source: `Superstore sales dataset.csv`, 9,994 order lines, no nulls, no duplicate rows, no ship-before-order rows.
- Order/Ship dates are day-first (`D/M/YYYY`) but inconsistently zero-padded; parsed accordingly rather than with a fixed format.
- All dollar figures are as recorded in the source data (Sales, Profit, Discount, Quantity); no currency conversion or inflation adjustment applied.
- Forecast built on monthly-aggregated Sales; Prophet (multiplicative seasonality) and auto-ARIMA/SARIMA were compared on an out-of-sample holdout before selecting Prophet for the production forecast.
- Full chart set, tables, and section-by-section detail behind every finding above are in `outputs/RESULTS.md`.